

PIR - Customers unable to access Basket Page

M&S Digital&Tech Post Incident Review



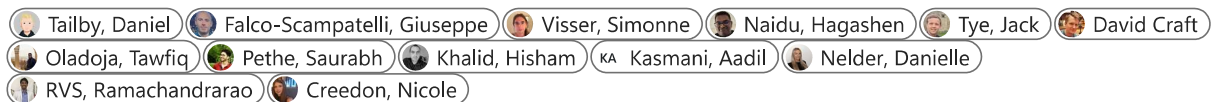
Incident Ref: 90659216

Incident Date: Wed, Mar 12, 2025

PIR Date: Thu, Mar 13, 2025

Author: Priyabrata Sahu

Attendees:



Technical Owner: Oladoja, Tawfiq

Analyses the root cause and provide technical recommendations to prevent incident recurrence.

Develop and oversee the implementation of an action plan to address the root causes.

Work closely with cross-functional teams to gather information and insights.

PIR Reviewer: Webster, Charly

Validates the root cause analysis, corrective actions, and preventive measures before signing off, ensuring alignment with business and operational standards.

1. Incident Summary :

Provide a concise overview of the incident with brief update on issue, impact, root cause and resolution.

Alerts have identified a significant surge in errors experienced by users attempting to access the basket page on both desktop and mobile browsers between 14:40 and 15:20. This technical issue affected approximately 1,000 customers, who faced challenges either in viewing their baskets or successfully completing the checkout process. Consequently, this situation resulted in a notably poor customer experience, raising concerns for any business striving to deliver seamless service.

An initial investigation into the issue has suggested that a recent change made within the price domain was likely the source of these errors. In response, this modification has been reverted, which should help restore normal service operations and resolve the issues customers were encountering.

2. Incident Chronology:

Include a record of the data/time of key events. e.g. "07:36 Incident started, notified by ..." **keep fact based".**

Wed, Mar 12, 2025

14:42 - Price domain made a change <https://github.com/DigitalInnovation/price/pull/13...>

14:48 - Basket alert raised for anomalous increase in error rate & acknowledged by Tailby, Daniel

14:53 - Second Basket alert raised for page views with errors & acknowledged by Tailby, Daniel

14:53 - Customer Order alert raised for failure of Synthetic tests and acknowledged by Shrivastav, Vikas

14:54 - Upon initial analysis, it was determined that the problem is coming from Price domain while fetching Quotes.

14:55 - Price domain notified by Pager Duty and acknowledged by  Falco-Scampatelli, Giuseppe

15:00 - Customer Order alert reassigned to Price domain

15:01 - Spike in errors reported in Tech Ops 

15:05 - Identified problem by looking at the logs in New Relic. This was caused by an endpoint change which wasn't whitelisted in Azure Front Door. The current smoke test didn't pick up the change and allowed the deployment. 

15:06 - Reverted the change by 

15:07 - Incident Channel notified 

15:10 - Initial investigation suggested on Price domain errors rather than Onyx Auth

15:15 - Incident bridge opened for issue/resolution progress and impact statement

15:25 - Incident loop created to capture more information

15:28 - Deployment completed and incident resolved 

15:50 - Based on the impact and customer experience, agreed to send Sev-1 incident comms

16:02 - Sev-1 incident comms shared to wider stakeholders

3. Incident Resolution

 *The actions taken to restore services and mitigate impact.*

Fix details

The price team rolled back the change that resulted in the incident.

4. Root Cause

 *Refers to the fundamental reason (s) why an incident occurred including problem description, analysis techniques, contributing factors, evidence with concise root cause statements*

Price team updated the URI of an endpoint but overlooked revising the list of allowed URIs. Additionally, they incorporated versioning by adding "v1.". The current smoke tests were incomplete and didn't detect this scenario and the build was successful.

5. Findings

 *Refer to the key insights and conclusions drawn from the incident analysis - what went well/what went wrong highlighting gaps, failures, or inefficiencies.*

5.1. Observability –

 *Evaluating monitoring, logging, and alerting effectiveness in detecting the incident.*

Multiple alerts raised to different downstream teams including Basket, Customer Order, Price. Business Ops reported on Content Square on high number 5xx errors.

Basket team were the first to receive an alert (As they experienced the front-end errors) and this was received via Pager Duty 6 minutes after the first errors occurred.

Price team (Owners of the change that caused the issue) received an alert 13 minutes after the first alert and using the logs were able to quickly identify that their change was causing the issue.

5.2. Technical Challenges -

 *Identifying system, infrastructure, or architectural issues that complicated resolution.*

Impact to Basket appeared to be around 20 - 25%. Just wanted to understand why the impact was not across all Baskets? This was confirmed in the call advising that the change was aimed specifically at Wishlist and so it is assumed that baskets with a Wishlist would have been impacted therefore confirming the reason why we did not see a complete failure of all baskets following the change.

The option to revert was preferred over fixing forward and is believed to have restored service within 10 minutes of initiation of the reversion

5.3. MIM involvement & impact assessment -

 *Timely engagement of MIM and analysing the impact on business operations based on top 10 business processes*

Gap in impact assessment.

Service Management team to review documentation on engaging with team via TechOps & Incident channels. (Updated on this section - *Technical Details and Useful Documents*)

5.4. Incident handling/ stakeholder management –

 *Reviewing how teams collaborated to drive the incident to resolution and engaged with key stakeholders.*

Price domain team was looking at the issue once an alert was received and following investigation began reverting the change that they believed caused the issue circa 15 minutes after alert was received.

5.5. Business Communication –

👉 *Clear, concise & timely communication to stakeholders, focusing on business impact and recovery actions.*

Incident communications were shared with Sev-1 stakeholders. The delay in disseminating these communications highlights the necessity for an improved impact framework and a more effective incident communications process.

5.6. Supplier Engagement –

👉 *Assessing the responsiveness and support provided by external vendors and internal product teams.*

The execution of this task was proficiently managed by our internal technical teams, thereby negating the necessity for any external suppliers.

Mitigation Actions:

👉 *Agree actions for each of the findings to prevent recurrence, strengthen system reliability & stability, and improve response efficiency for future incidents*

	☰ Action	☰ Owner	📅 Due Date	☰ Remarks	🏷️ Action Category
1	Enhance the Smoke Test to identify issues when modifying the endpoint URI.		Wed, Mar 19, 2025		Problem Avoidance
2	Incorporate the allowed_uri in Terraform with the updated endpoint_uri , and verify that the Smoke Test passes successfully.		Wed, Mar 19, 2025		Problem Avoidance
3	Assess the alerts to determine if the price domain requires earlier notifications.		Wed, Mar 19, 2025		Problem Avoidance
4	An evaluation of the change management process for standard pipelines, focusing on integration with Remedy, is essential for optimizing workflow and ensuring seamless operations.		Fri, Mar 28, 2025		Impact assessment
5	Until E2E test script is fixed, all production changes will be manually reviewed and released with caution. This action should have a date as well		Fri, Mar 28, 2025		Problem Avoidance

Please refer to  Major_ Significant and Key Incident Tracker v1.0.xlsx for progress updates on Mitigations

Technical Details and Useful Documents:

What's the purpose of using Incident Channel? [Link](#)

Reporting of all Severity-1 or Severity-2 incidents impacting technology

Purpose: This channel is dedicated to posting and tracking Significant Incidents or higher within our organisation. It serves as a central hub for real-time updates, impact assessments, and resolution steps for incidents that significantly affect our operations, customers, or services.

When should team use TechOps Channel? [Link](#)

When severity is not clear or when business teams report an issue that requires addressing by technology.

Purpose: To discuss any operational issue in the production environment that requires investigation.